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MGT 663

Strategic Opportunities in Impact Enterprise

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1. Introduction

Inspired by the world-renowned inventor and electrical engineer Nikola Tesla, an electric car start-up set out with one purpose— to accelerate the world’s transition to sustainable energy (1. Tesla, 2020). The question then became “how?” Previous electric car ventures had failed from automobile giants, such as General Motor’s EV-1, considered at the time to be the best vehicle of its kind on the market. The company had invested over one billion dollars in the car’s development and production facilities, yet broad market appeal and buy in never followed suit. Despite avid support from technology enthusiasts and environmentalists, the product line was discontinued and later featured in a 2006 documentary “Who Killed the Electric Car” (2. Baer, 2014).

With the automobile industry disavowing electric vehicles, coupled with the technological advancement of lithium-ion battery power density and reduction in production costs, Silicon Valley engineers Marc Tarpenning and Martin Eberhard identified an opportunity in the market. In 2003, Tesla Motors was founded by the two entrepreneurs, setting out on their first mission of creating the world’s first electric sports car (3. Thompson, 2020). The car would not only be the first of its kind, but boast superior performance, function, and value compared to traditional sports cars, completely redefining the perception of electric vehicles. In 2004, Tarpenning and Eberhard pitched the idea to Elon Musk and a partnership was formed (Baer, 2014).

Supported by several rounds of funding spearheaded by Musk, the Roadster began development. Rumblings of this bold vision spread throughout Silicon Valley and in 2006 the Roadster made its debut at an event in Santa Monica, CA. Stunning spectators, 127 vehicles were pre-ordered for \$100,000, with each customer dubbed a "Signature One Hundred Member" (Baer, 2014). Later that year, Musk penned a blog post “The Secret Tesla Motors Master Plan (just between you and me)” (4. Musk, 2006) outlining his vision for the company beyond the Roadster:

“Build sports car
Use that money to build an affordable car
Use that money to build an even more affordable car
While doing above, also provide zero emission electric power generation options”

Despite public excitement and momentum, the pre-ordered Roadsters incurred several production delays before being released in mid March of 2008. During this same period, Tesla experienced organizational restructuring due to internal conflicts, with Eberhard leaving the company and Musk later becoming the CEO in October of 2008 (Baer, 2014). With Musk now at the helm, he navigated the company through the 2008 financial crisis, narrowly avoiding bankruptcy with a \$40 million financing round (Thompson, 2020).

Since then, Tesla has been unstoppable. In 2009 the first electric sedan, the Model S, was announced and received over 1,000 preorders. The following year, Tesla became the first American automobile company to IPO since Ford Motor in 1956, raising over \$226 million, and opened its first factory in Fremont, California. With the benefits of additional funding, the Model X SUV prototype was announced in 2012, increasing traffic to the company’s website by 2,800% and \$40 million in preorders (Thompson, 2020).

Soaring in popularity and committed to his promise of affordable and accessible sustainable transportation, Elon open-sourced his patents in 2014. That same year, Tesla announced the creation of a battery factory, referred to as a Gigafactory, in Nevada with the goal of reducing battery prices by 30% in an effort to make future models more affordable (5. Thompson, 2020). By 2016, Musk released the prototype for the Model 3, and classified it as the company's first mass-market car. Months later, the ambitious CEO penned "Master Plan, Part Deux" outlining his newfound goals:

1. Develop "stunning" solar roofs that seamlessly integrate with Tesla's battery storage.
2. Roll out more affordable vehicles "to address all major segments."
3. Advance its self-driving technology so that it is "ten times safer" than manual driving.
4. Roll out a car-sharing program that enables Tesla owners to make money by renting out their autonomous car."

Wanting to signify the organization's expansion into the solar business, the official name was changed from Tesla Motors to Tesla Inc in 2017 after the purchase of SolarCity, a solar panel company. Since then, a wide range of energy products have been announced, such as the Solar Roof, along with new vehicle models (Thompson, 2020). New models include: the Tesla Semi, a new version of the Tesla Roadster, a compact SUV named Model Y, and the Tesla Cybertruck. Most recently, the Tesla Model S Long Range Plus is the first electric vehicle to achieve an EPA-rated range of 402 miles (Tesla, 2020).

To accompany the growth and global demand of Tesla products, a second Gigafactory was built in Buffalo, New York to produce solar cells and energy storage products. A third Gigafactory in Shanghai to cater to the Chinese market, a fourth in Berlin to focus on the manufacturing of the Model Y, and a fifth in Austin, Texas to produce Model 3, Model Y, the Tesla Semi, and Tesla Cybertruck (Thompson, 2020). Potential future locations have been kept secret.

As of 2021, Tesla shows no signs of stopping. In June, the Model 3 became the best selling electric car worldwide with over 1 million sold. Currently valued at \$1 trillion, Tesla is continuing to support the world's transition to sustainable energy without compromising quality, affordability, and convenience (6. Lambert, 2021). It leads one to wonder "what's next?"

2. Industry Analysis

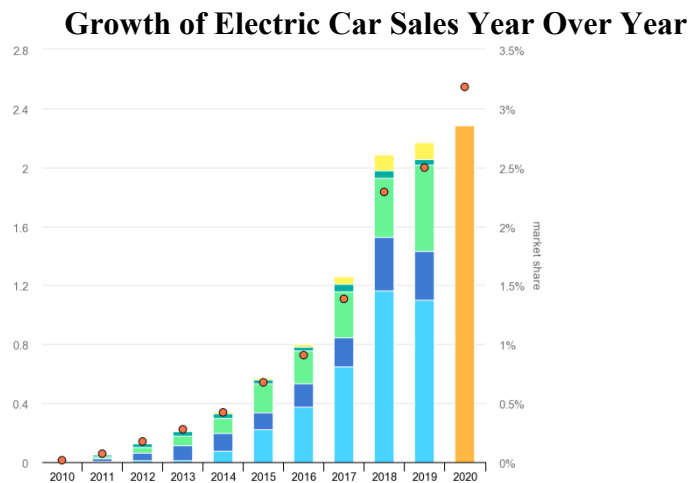
2.1 Sector Definition

The industry that Tesla operates in broadly is the sustainable transportation industry. This includes trains, hybrid cars, and low-carbon air-travel. It operates more specifically in the automotive/electric vehicle space. Tesla's operations also include energy generation (through SolarCity) as well as storage, through its production and development of Lithium Ion Batteries (1. Yahoo Finance, 2021). The aspect of this sector we will look at will be the automotive industry, specifically electric vehicles. Tesla is fairly vertically integrated in that they construct most of their vehicles and source some of their products, specifically lithium-ion batteries, from themselves. Tesla also sells direct to consumers and provides service to vehicles at company owned and operated service centers and service locations. Tesla is currently building 100 new service centers which will bring its total to 250, and is rolling out 350 on demand service vans (2. Dudovskiy, 2019).

2.2 Sector History, Growth and Current Status

The electric vehicle industry started around the late 1800's with the first successful electric car being sold in 1890. It was created by a chemist and could travel at 14 miles an hour with 6 passengers. Electric vehicles reached their heyday from 1908 to 1912 with a third of all vehicles on the road being electric. With the introduction of the Model T in 1908 the early electric vehicle market started to decline and was all but gone by 1935, due to cheap gas and an improved road system connecting cities, requiring longer range vehicles (3. Energy.gov, n.d.). Electric cars were for the most part irrelevant until around the 1990's. Government regulation renewed an interest and GM produced "EV 1", which gained a cult following, but not much more than that. It had a top speed of 80 miles per hour and a range of 90 miles (4. Winfield, 1997). This brings us into the 2000's, specifically 2003, when Tesla Motors was founded. From 2009 to 2015 the Department of Energy invested in charging infrastructure and batteries, reducing the cost by 50 percent. In this timeframe, GM released the Chevy Volt which is the first plug-in charge car, and Nissan released the LEAF, an all electric car. (Energy.gov, n.d.).

More recently, however, electric cars have begun to take off in earnest. With the development of high capacity, cheap batteries and a growing awareness of climate change, electric vehicles are beginning to outcompete standard gas powered cars in some countries, and look poised to replace all sales somewhere in the 2030's. Electric vehicles sales in 2020 grew about 40 percent from 2019, even as sales of regular cars slumped during the COVID pandemic. They accounted for around three percent of new car sales in 2020 (8. IEA, 2020). When looking at the curve of EV sales, it appears to be exponential (5. Adler, 2021). In fact, many car manufacturers including Jaguar, Volvo, and Hyundai have committed to not sell anything but EV's by 2030 (6. Duffy, 2021) with GM committing to a fully electric offering by 2035 (7. Boudette, 2021).



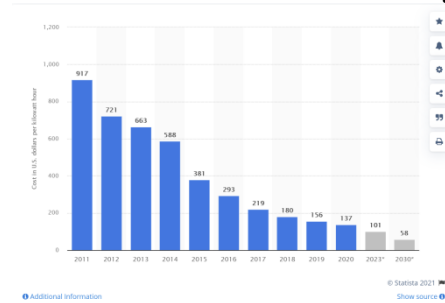
(8. IEA, 2020)

2.3 Key Technologies

The key technologies driving EV uptake are cheaper, more powerful batteries. Lithium-ion costs have plummeted to less than \$200 for 137 Kw/hr in 2020 which has driven the cost of EVs down significantly (10. Statistica, 2021). Fast charging is another aspect that has helped adoption—a vehicle can add 100

Miles of range in less than 30 minutes (9. Pod Point, 2021). Autopilot is another key technology, but only applies to Tesla as they are the only company currently that has full autopilot capabilities.

Cost of Lithium Ion Battery Year to Year



(10. Statista, 2021)

2.4 Value Chain Analysis

See Figure 7.

Inbound Logistics: Tesla buys the raw materials necessary to build EVs, including but not limited to: carbon steel, aluminum, resins, glass, lithium, nickel, cobalt, and copper. The company partnered with Panasonic to create the Nevada Gigafactory in order to produce their own batteries.

Operations: Tesla designs and assembles vehicles at its facilities in California, the Netherlands, and China. Because they own the battery technology, they can smooth out the production and logistics processes for manufacturing rather than outsourcing.

Outbound Logistics: By forgoing the model of producing cars and stocking them at a dealership, Tesla is able to reduce costs because customers build their car to order and the company sells directly to end users. This is a significant source of value creation for Tesla because direct sales decrease the time of delivery to customers and keep the prices of Tesla's EVs from being even higher.

Marketing and Sales: Tesla has utilized online sales and "Apple-like" retail outlets in heavily populated areas to increase brand exposure. "Zero Dollar Marketing" has also been at play. Tesla attracts extensive media coverage without paying for advertising, thus increasing awareness and sales without having to staff a large marketing department.

Service: Currently, the Supercharger Network is only available to Tesla owners, and enables them to quickly charge their cars for free while they are on the road. The Supercharger network is located along main routes and near other amenities so Tesla owners can charge their EV while at a restaurant, shopping, or at work.

Firm Infrastructure: Tesla has two arms of production: automotive (design, manufacturing, and sale of EVs) and energy generation (design, manufacturing, installation, and sale of solar energy generation and storage products).

Human Resource Management: There are some concerns, which will be touched upon in the Impact Analysis of this paper, but Tesla states their commitment to facilitating a safe and cohesive workplace

culture in their Corporate Social Responsibility report. “Tesla is on a mission to accelerate the world’s transition to sustainable energy. We focus relentlessly on innovation and rely on collaboration across our teams to help achieve this goal. Along the way, we are committed to maintaining safe and humane working conditions, ensuring that all workers, including those within our supply chain, are treated with respect and dignity and that our manufacturing processes are environmentally responsible. Internally, we are continuing to build a culture that is safe, respectful, fair and inclusive for all our employees.” (11. Tesla, 2020)

Technology Development: Tesla’s proprietary powertrain, lithium-ion battery cells, radar sensors, computer system, autopilot control systems, and solar energy products are all sources of major differentiation for the company. Tesla is more like a tech company on wheels than a traditional automaker.

2.5 Five Forces Analysis

Industry Competition: HIGH

- There are few competitive firms (currently), but they are some of the most established carmakers in the world and are highly innovative and motivated to take market share
- GM has an experience advantage as a traditional auto company in the US market and one of the oldest car makers in the world.
- Volkswagen, another well known and experienced automaker, predicts that 50% of its sales will be EVs by 2030. VW has made huge investments in their EV technologies recently.
- Foreign automakers, like Nio, are especially competitive with Tesla in China.

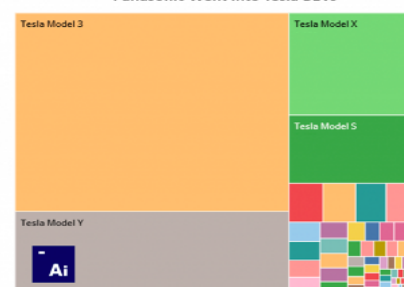
Threat of new entrants: LOW

- Barriers to entry are high:
 - High cost to create a new brand
 - High economies of scale
 - Incumbent advantages: who is going to buy a car from a new company they’ve never heard of?
 - Incumbent retaliation: how will incumbents like Tesla, GM and Volkswagen market against new entrants?
- Despite barriers to entry, the electric car market is starting to grow substantially, and EV sales are expected to be 38% of all vehicles sold by 2040 (12. Wood Mackenzie, 2019).

Power of suppliers: MODERATE

- Tesla builds the base of its cars, but relies on suppliers for ancillary parts like windshields, brakes, etc.
- Panasonic: lithium-ion cell supplier for all Tesla models which would indicate huge power, but Tesla accounts for almost 90% of Panasonic’s xEV battery sales.
 - Figure: (13. Adamas Intelligence, 2021)

Nearly 90% of Passenger EV Battery Capacity Deployed Onto Roads by Panasonic Went Into Tesla BEVs



Power of customers: HIGH

- Switching costs—what's stopping someone from buying a Volkswagen ID.4 SUV for roughly the same base price and more range than a Tesla Model 3 sedan?
- Differentiation—As more EVs come to market, Tesla may lose its grip on the majority share of the EV market if there are more offerings at a similar price with equal/better range and more capability that customers begin to gravitate toward.

Threat of substitute products: LOW

- Hydrogen Fuel cell cars are an option for cleaner transportation that do not require new charging infrastructure or battery technology. Hydrogen can ostensibly be pumped using an existing network of gas stations. However, FCVs are less energy efficient overall when looking at input vs output of power: there is a 20% loss of power from charge to drive for EVs vs. a 62% loss of power from electrolysis, transport, conversion to electricity, and drive for FCVs (14. Morris, 2020)
- No other EV manufacturers in the US are coming close to Tesla's sales right now (see Competitor Analysis)

2.6 Industry Drivers and Trends

EVs in Regional markets: Markets for battery electric vehicles (BEVs) and plug-in hybrid electric vehicles (PHEVs) have started to build up momentum worldwide. Before COVID-19, the combined annual sales of EVs and PHEVs tipped over two million for the first time in 2019 and reached 2.5% share of all new cars. BEVs account for 74% of global EV sales with an increase of 6% points since 2018. Reaching these numbers required a substantial investment from OEMs to develop technology, R&D, and factory redesign. Governments have also played an important role in pushing the supply curve upward, especially in Europe, where stricter emissions standards have been imposed on manufacturers to increase the production of zero-emission vehicles. (15. Deloitte, n.d.) See Figure 1.

- *US:* According to the International Energy Agency (IEA), the US represents only about 17% of the world's total stock of 10.2 million EVs. Since 2018, EVs account for about 2% of the US new car market. Sales growth from previous years was drawn down during the pandemic, selling 50% fewer PHEVs compared to 2018 and EV sales dropped 3.2% from 2018. (16. DeSilver, 2021) See Figure 2.
- *China:* China is the largest EV market by far, with 44% of the world's total stock. Since 2014, the EV market share in China has increased year by year, reaching 5.7% of the total new car market. Sales of EVs grew by 17% from 2018 to 2019. Due to COVID-19, China's EV market fell 56% in Q1 2020. (DeSilver, 2021) See Figure 2.
- *Europe:* The European Union has the fastest growing market for EVs in the world, with a 93% growth from 2018 to 2019. The EU accounts for 31% of the world's total stock with nearly 3.2 million EVs. In general, Nordic nations and Western Europe possess the highest market shares, reaching their highest levels since 2018. In countries like Norway and Iceland, the EV market represents more than 50% of the total new car sales. (DeSilver, 2021)
- *Rest of the world:* Other markets around the world have remained stagnant due to various factors such as lack of government commitment to EVs, insufficient or unsuitable infrastructure, and economic and cultural differences regarding mobility models. These markets represent nearly 8% of total EV sales. Canada, New Zealand, and South Korea have low EV penetration; however,

these countries have started to capture a small share of the internal combustion engine (ICE) market. (DeSilver, 2021)

Growth Drivers:

- *Changing consumer sentiment:* Consumers are game-changers for the future of EVs. In a Deloitte Global Auto Consumer Study from 2020, consumers expressed their change in concern regarding EVs in terms of safety, accessibility, functionality, and price. From 2018 to 2020, consumer attitudes toward EVs have changed in different regional markets. For example, concerns over cost/price for premium cars have diminished in the US and EU, but not in China. Lack of charging infrastructure has become the top priority for consumers in almost every country. (Deloitte, n.d.) See Figure 4
- *Policy and legislation:* Government intervention can accelerate EV adoption and sales growth. The EU is taking the lead in undertaking net-zero emissions policies and supporting a transition to EVs, most notably in Norway and the Netherlands. Among the different range of policies, the most effective in encouraging the growth of EV adoption are fuel economy and emission targets, city access restrictions, and financial incentives. (Deloitte, n.d.)
- *OEM vehicle strategy:* OEM's EV strategy for the next decade seems quite ambitious. Several industry players have put their bets on EVs and reduced prices for consumers. According to statistics cited by the European Federation for Transport and Environment, Europe should expect over 100 new EV models available by 2023. In the United States, IHS Markit predicts there will be 130 available models by 2026, offered by 43 brands. In the short term, subsidies and increased efficiency of EVs are key to achieving parity with ICEs and speeding up adoption among consumers. (Deloitte, n.d.)

2.7 Recent Developments



Automotive:

- *Full Self-Driving:* Full Self-Driving (FSD) is the new standard for Tesla. Their vision is to establish an autonomous Tesla ride-hailing network in the near future. Currently, the company is offering in their vehicles certain advanced driver assist systems under our Autopilot and FSD options. (17. Tesla, 2020)
- *Vehicle Control and Infotainment Software:* New software capabilities allowed Tesla to improve customers' experience. Performance and safety systems are controlled by sophisticated built-in software. Control systems in EVs optimize performance, customize vehicle behavior, manage to charge, and control all infotainment functions. (Tesla, 2020) Additionally, advanced connectivity solutions have become increasingly popular in EVs. For example, personalized contextual advertising based on driving routes provides relevant information to drivers and reduces time and energy consumption on each route. (18. McKinsey, 2021)

Energy storage and Powertrain: Tesla's core vehicle strategy relies on its batteries to optimize energy consumption and increase performance. To enable a greater supply of cells for their products with higher energy density at lower costs, Tesla is currently using their expertise to develop a new proprietary lithium-ion battery cell and improve manufacturing processes. (Tesla, 2020)

Powertrain engineering has allowed Tesla to develop powertrain systems that are adaptable, efficient, reliable, and cost-effective while withstanding the rigors of the automotive environment. The company is offering dual-motor powertrain vehicles, which use two electric motors to maximize traction, and just recently, they have introduced a vehicle powertrain technology featuring three electric motors for further increased performance. (Tesla, 2020)

2.8 Key Success Factors

Environmental Impact: Reducing their carbon footprint is inherent to Tesla's growth. Tesla has conducted a Life Cycle Analysis which includes most of the Scope 1, 2, and 3 emissions, including the vehicle manufacturing phase, emissions from their supply chain, vehicle use, and end-of-life. (19. Tesla, 2020)

Product Impact: Tesla has developed an EV beyond industry standards that is pushing the market forward. The company is focused on developing products that are not only sustainable but also superior to fossil-fuel alternatives in every way. Tesla vehicles provide increased performance, safety, efficiency, and competitive prices compared to other premium cars in the market. (Tesla, 2020)

Supply Chain: Tesla suppliers are required to provide evidence of management systems that ensure social, environmental, and sustainability best practices in their operations, as well as to demonstrate a commitment to responsible sourcing into their supply chains. (Tesla, 2020)

People and Culture: Employees play a critical role in achieving Tesla's sustainability and productivity goals. At the center of their People strategy is maintaining meaningful engagement with employees, robust culture of safety, clear development pathways for all employees, highly competitive compensation programs, and principles of diversity, equity, and inclusion built into all aspects of hiring, promotion, and retention. (Tesla, 2020)

3. Strategic Analysis

3.1 Mission, Purpose, and Values

Purpose: Referenced in the 2020 Impact Report, the purpose of Tesla is to accelerate the world's transition to sustainable energy without compromising quality and convenience.

Mission: Tesla has had several missions throughout the life of the company. Outlined in both of Elon Musk's "Master Plans," he lists four specific but intertwined missions for the organization. In the first master plan, "The Secret Tesla Motors Master Plan (just between you and me)," Musk states the mission is to build a sports car, use the earnings to invest in a more affordable car, and to use those earnings to invest in an even more affordable car. While doing so, Tesla will create sources of zero emissions electricity.

A decade later, Musk declared that Tesla was close to completing the first set of missions. A new plan titled "Master Plan, Part Deux" outlined the next four. The company would now be focusing on creating solar roofs that integrate with battery storage, expanding the EV product line, developing self-driving capabilities that are 10 times safer than manual driving, and using your Tesla as a way to earn extra income.

According to Tesla's 2020 Impact Report, these are enmeshed into the overarching mission of creating a "complete energy and transportation ecosystem" that is vertically integrated.

Values: Sustainability is at the core of Tesla and influences the company's everyday decisions. Other values evident throughout the company include the emphasis on quality and the drive for innovation.

3.2 Business Units and Products

Tesla focuses on the development and production of electric vehicles powered by lithium ion batteries.

Product Offerings:

- Model S: a high performing sedan that boasts the longest driving range and the fastest acceleration of any electric vehicle on the market.
- Model 3: a high performing sedan at a more affordable price.
- Model X: a high performing SUV that seats up to seven people. This model also has a large amount of storage space and the highest towing capacity of any electric SUV.
- Model Y: an all-wheel drive SUV that can seat seven people and is capable of off-roading adventures.
- Full Self-Driving Capability: features include automated parallel or perpendicular parking, lane changes, and the summoning of the vehicle to find its user in the parking lot.
- Autopilot: assists the driver in steering within a lane and senses the speed of traffic to slow down or accelerate.

Future Offerings:

- Semi: with a maximum range of 500 miles per charge, this model will have four independent motors and autopilot. It is estimated to save semi truck drivers \$200,000 a year in fuel.
- Cybertruck: made with a durable and impenetrable glass and stainless steel to boost the safety of passengers as well as to prevent the wear and tear of the vehicle.
- Roadster: a high performance and aerodynamic sports car that can become a convertible and boasts a driving range of 620 miles.

Tesla Energy is a subsidiary of Tesla. This portion of the company focuses on the development, production, and installation of solar panels and batteries with the purpose of creating and storing clean energy.

Product Offerings:

- Solar Roof: solar roof tiles that cover the entirety of one's roof. These tiles are three times stronger than standard roofing tiles and produce solar energy to power one's home.
- Solar Panels: durable and sleek panels placed on one's roof to generate a portion of solar energy to power one's home.
- Powerwall: a rechargeable battery that stores unused solar energy for later use.

- Megapack and Powerpack: large scale rechargeable batteries used for commercial or utility purposes.
- Autobidder: an automatic trading and control platform that evaluates markets in real time in order to monetize battery assets, such as stored solar energy.

3.3 Primary Target Market

Tesla's primary target market is all drivers of vehicles worldwide, as well as those who need electric power. The plan that they have followed was outlined by Elon Musk in 2006 (1. Musk, 2006). The plan stated that Tesla would create a premium electric sports vehicle to sell, then use all of the cash generated from that vehicle to produce R&D and production capability for a cheaper four-door electric vehicle. Then they would use that money to create more models, and help with the transition to renewable energy. This strategy has worked quite well—Tesla now has four models of vehicles with plans to make more, including a consumer truck (Cybertruck), and is slated to release a Semi-truck in 2022 (2. Levin, 2021). Tesla also installs charging stations across the U.S., produced 500,000 cars in 2020, and is on track to produce 1 million in 2021 (3. Iqtidar, 2021). These are impressive numbers given that Tesla vehicles are made to order.

3.4 Value Proposition to the Customer

Tesla's value proposition is to deliver a green automotive alternative that outperforms the standard, has advanced capabilities and updates, is sleek and cool, has superior functionality, and includes low cost or free charging. Tesla also equips homes with battery packs and solar roofs to further customers investment in the electrified economy.

3.5 Competitor Analysis

Q2 2021 EV Sales (4. Cox Automotive, 2021):

ELECTRIC VEHICLES								
	Q2 Sales			YTD Sales			Segment Share	
	2021	2020	YOY	2,021	2,020	YOY	Q2 2021	YTD
Tesla Model Y	38,877	1,000	3788%	76,577	1,000	7558%	32.9%	35.3%
Tesla Model 3	26,680	20,300	31%	47,380	68,300	-57%	22.6%	21.8%
Chevy Bolt EV/EUV	11,263	2,498	351%	20,288	8,371	54%	9.5%	9.4%
Mustang Mach-E	6,361	-	-	12,975	-	-	5.4%	6.0%
Tesla Model X	6,098	1,850	230%	12,198	5,450	69%	5.2%	5.6%
VW ID.4	5,756	-	-	6,230	-	-	4.9%	2.9%
Nissan Leaf	4,804	1,049	358%	7,729	3,007	49%	4.1%	3.6%
Tesla Model S	4,575	2,350	95%	9,375	6,950	4%	3.9%	4.3%
Porsche Taycan	3,359	818	311%	5,367	1,039	809%	2.8%	2.5%
Audi e-tron	2,560	1,161	120%	6,884	2,872	153%	2.2%	3.2%
Volvo XC40	2,009	-	-	2,329	-	-	1.7%	1.1%
Hyundai Kona	1,845	623	196%	3,401	1,387	145%	1.6%	1.6%
Kia Niro	1,726	492	251%	2,406	1,089	121%	1.5%	1.1%
Hyundai Ioniq	830	134	519%	1,275	287	344%	0.7%	0.6%
BMW i3	511	188	172%	851	447	90%	0.4%	0.4%
Mini Cooper	431	73	490%	738	99	645%	0.4%	0.3%
Jaguar I-Pace	404	776	-48%	655	1,154	-34%	0.3%	0.3%
Polestar2	144	-	-	259	-	-	0.1%	0.1%
Total EV (Estimates)	118,233	33,312	255%	216,917	101,452	114%	100.0%	100.0%

In the current EV space, Tesla's top US competitors are:

- Chevrolet (GM): The Bolt EV (259 mile range, starting at \$31,000) is the closest competitor to Tesla in the EV market in terms of units sold. The interior is less tech-oriented (there is a touch screen, but many controls operate in a familiar way), and not as luxurious as Tesla's interiors.

The Bolt does have some appeal to consumers who don't want all the technology and gadgetry but want an EV that performs well. Charging options include a Level 1 charger that can plug in to a standard 120-Volt outlet (4 miles of range per hour of charging), a Level 2 charger that requires a 240-Volt outlet to be installed in your home (26 miles of range per hour of charging), and a network of 40,000+ public charging stations along with an app called Energy Assist that will help you plan routes and give you an idea of how long you'll need to charge up along the way. Chevy is working on some concept vehicles that are blending classic styles with electric performance to create a new perspective in the EV market (5. GM, n.d.).

- Ford: The Mustang Mach-E (305 mile range, starting at \$42,895) is Tesla's next closest competitor for units sold in the US EV market. One touch screen facilitates all center console controls, similar to like Tesla's style, but some reviews note that the interior is not the quality that you would find in a Tesla. All-wheel drive is an optional package on the Mach-E as well, appealing to drivers who commute in the snow. For charging, Ford has the Connected Charge Station: installation of a 240-Volt outlet in your home can provide 28 miles per hour of charging, but a standard 120-Volt outlet provides about 3 miles per hour of charge. The Blue Oval Charging Network has over 19,500 charging stations but does have some stations that charge money for charging access. The FordPass App allows you to find charging stations near you and plan trip routes according to where you'll be able to charge up.
- VW: The ID.4's the pricing and range is comparable to Tesla's Model 3, starting at \$39,995 with 260 miles of range. VW is taking a page from Tesla's book and making the ID.4 to order and will ship the car to a dealer once it is ready.

Other competitor notes:

The VW Group has plans to launch 70 EV options by 2030, with a goal of producing 1.5 million units by 2025. They are building six 40GW battery factories in Europe (Tesla's Nevada Gigafactory has a 35GW capacity), which will be operating by 2030, with plans to make batteries cheaper and out of recycled materials (6. Beresford, 2021). The company also has plans to build more fast charging stations in North America to expand the current network.

Ford is currently working on producing electric trucks to compete with Tesla's Cybertruck. The F-150 Lightning has 130,000 reservations, according to recent stakeholder news updates. According to Elon Musk, Tesla stopped counting Cybertruck reservations after they hit 600,000. However, consumers surveyed stated that they were more likely to consider Ford's electric truck over Tesla's based on price, performance, and design (7. Cox Automotive, 2021).

Nio, a premium EV carmaker in China, is one of Tesla's biggest competitors outside of the US. The company's CEO has stated "we want to provide better product and service at prices lower than Tesla Inc's" (8. Thompson, 2021). While they are a relatively new company (founded in 2014, but manufacturing only started in 2017), an initial quality survey by JD Power ranked Nio above Tesla in battery electric vehicles (9. JD Power, 2020). In addition, the value offerings to Nio consumers are unique. Their "Battery as a Service" subscription model allows Nio owners to swap batteries in order to avoid stopping to charge when covering long distances (up to 6 times per month). Owners also have access to Power Mobile: a van with a charging port that will come to your location to give you a quick charge (up to 15 times per month).

3.6 Source of Differentiation Relative to Customers

Distribution network:

- Direct Sales: vehicle sales channels currently include a website and an international network of company-owned stores.
- Used vehicle sales: used vehicle business supports new vehicle sales by integrating the trade-in of a customer's existing Tesla or non-Tesla vehicle with the sale of a new or used Tesla vehicle. (10. Tesla, 2020)

Service:

- Public charging: Tesla has a growing global network of Tesla Superchargers, which are industrial grade, high-speed vehicle chargers. Supercharger stations are typically placed along well-traveled routes and in and around dense city centers.
- Tesla mobile service: this is a solution for electric vehicles at company-owned service locations and through Tesla Mobile Service technicians who perform work remotely at customers' homes or other locations.
- Vehicle Limited Warranties and Extended Service Plans: this service provide a manufacturer's limited warranty on all new and used Tesla vehicles sold, which may include separate limited warranties on certain components, specific types of damage or battery capacity retention. (Tesla, 2020)

Innovation:

- In-app upgrades: Tesla vehicles are capable of being updated remotely over-the-air. Customers may purchase additional paid options and features through the Tesla app. (Tesla, 2020)

3.7 Source of Competitive Advantage and Sustainability.

Technology: Tesla is more than an EV manufacturer. Their production processes go beyond the standards in the automobile industry. The company is focused on managing a reliable and sustainable ecosystem that makes its entire supply chain more resilient. Tesla expects to outcompete major EV players through continued progress on Autopilot, FSD and neural network capabilities, the Supercharger network, and infotainment offerings. (Tesla, 2020)

Intellectual Property: Tremendous efforts have been made to develop technologies and products that add intrinsic value and uniqueness to Tesla's portfolio. The strategic development of the company relies on these innovations and designs; therefore, Tesla has protected underlying intellectual property rights with respect to patents, trademarks, copyrights, trade secrets, and other measures, including through employee and third-party nondisclosure agreements and other contractual arrangements. (Tesla, 2020)

Human capital resources: Tesla's skilled and diverse labor force contributes to the company's growth and continuous development. The company has implemented programs, and practices to further enhance its employees' capabilities and attract the best talent. Tesla promotes a working environment that allows its employees to develop professionally and engage in a diverse multicultural network. (Tesla, 2020)

3.8 Core Competencies that Lead to Competitive Advantage

Dedicated Infrastructure: Tesla's dedicated infrastructure to EV production and energy storage assembly has allowed the company to improve its quality management processes and increase productivity rates.

Ceaseless Innovation: Over the years, Tesla has continuously evolved and improved its EV software, increased energy efficiency, and upgraded the customer experience.

Consistent high quality: Delivering next-generation EVs has been the standard for Tesla, from sourcing socially- and environmentally-friendly raw materials to ensuring each unit produced meets the highest industry standards in terms of design, manufacture, safety, and customer satisfaction.

3.9 Major Opportunities/Weaknesses/Threats

Opportunities:

- Continue partnerships with energy companies to push their battery and charging technology forwards.
- Expand the charging network across the US to provide more access to EVs in rural areas.
- Focus on the sustainability angle of EVs, and try to combine EV sales with solar roofing and storage sales.
- Elon Musk as the face of Tesla.

Weaknesses:

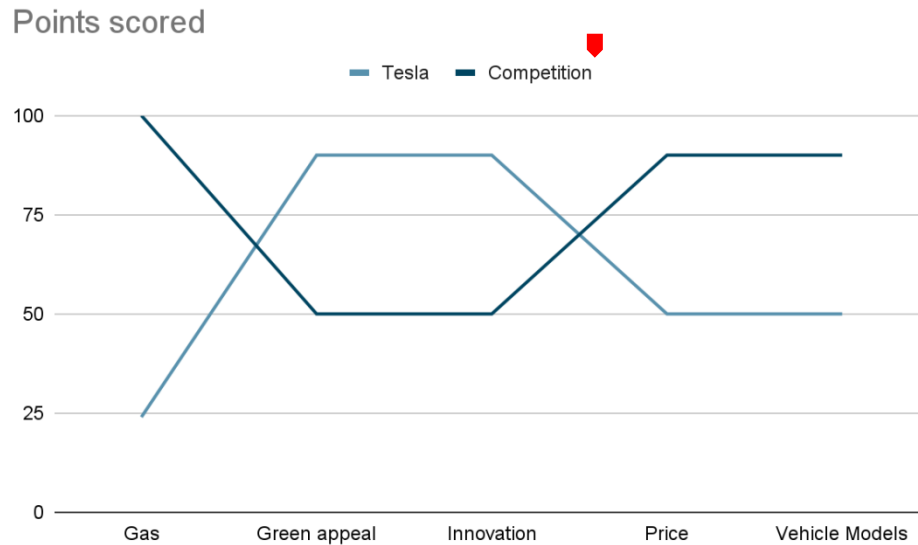
- High operational costs threaten profitability on an annual basis.
- Small target market of consumers who can afford and want to buy a luxury electric vehicle.
- Accessibility—consumers in rural areas don't have as much access to public EV charging, so they are not as likely to buy an EV.
- Elon Musk as the face of Tesla.

Threats:

- Competitors with cheaper EVs—if someone can buy an EV with similar range and safety offerings, and they don't care about the brand name, it's unlikely that they'll pay more for a Tesla.
- Battery technologies—if Tesla can't retain their place as the premier battery producer, they could be in trouble since it will be difficult to switch from their proprietary technology quickly.
- Customer demand for long-term sustainability—what will battery components be used for after the battery is no longer useful? Can Tesla EVs have their batteries replaced so the body can continue to be used indefinitely?
- Elon Musk as the face of Tesla.

3.10 Blue Ocean Analysis

Tesla is a poster child for Blue Ocean Analysis. They took the traditional model of transportation in the U.S., asked “What can we eliminate?” (fossil fuels), and raised the standards on almost every other category of competition, except price initially. Tesla created the green sports car industry. It then took a majority share of electric vehicle sales and was the first mover, essentially remaining uncontested by any other major automakers until recently. Tesla pays for charging, so customers essentially have free ‘gas’. Tesla eliminated dealerships as well, choosing instead to sell directly to the consumer and take care of long term maintenance. Tesla also outcompeted on safety, initially they were given 5 stars on the car safety tests and if the scoring went above 5 stars they would score 5.4 (11. Hollister, 2013).



Four actions framework:

<u>Eliminate</u> ● Fossil Fuels ● Pure Green play ● Driving ● Stigma around EVs	<u>Reduce</u> ● Diversity of vehicle models
<u>Raise</u> ● Speed and performance ● Convenience (self-driving) ● Style ● Customer care	<u>Create</u> ● Innovative onboard features (all screen display) ● Entirely new business model based around d2c sale, service and performance

4. Impact Analysis

4.1 Environmental Impact

As stated in Tesla's 2020 Impact Report, sustainability is at the heart of each of the company's decisions, missions, and innovations. Focused on transportation as well as energy production and storage, two of the largest sources of greenhouse gas emissions, Tesla continues to work towards viable and reliable zero-emissions alternatives through solar power and electric vehicles.

There are several positive environmental impacts associated with Tesla's supply chain design, production process and product offerings relayed in their 2020 Impact Report.

Supply Chain and Manufacturing Facility Localization: The expansion of production facilities to major markets in China and Europe reduces the emissions associated with shipping Tesla products solely from Fremont, California. A new US based factory in Austin, Texas will service the Eastern United States. As a result, the lifecycle emissions of models such as the Model 3 are reduced. An emphasis on car parts manufactured close to existing facilities as well as producing battery packs and seating in-house are further reducing emissions associated with transporting the necessary components for production.

Renewable Energy: Since 2012, renewable energy generation from solar panels installed on the rooftops of the production facilities has outpaced usage eightfold. Excess energy is sold to the grid, increasing the share of renewables within the electricity mix and reducing the need for fossil fuel powered sources.

Air Pollution Reduction: The burning of fossil fuels releases toxic air pollutants such as NO_x and particulate matter. As a result, 8 million premature deaths a year are associated with the pollution from combustion. Tesla eliminates the problem entirely as all of the products are battery powered.

Battery Recycling and Development: To reduce the need for mined raw materials, Tesla is in the finishing stages of infrastructure for recycling end of life batteries. The organization is also developing battery technology that eliminates the need for cobalt, one of the three common metals in batteries. Finally, Tesla has already achieved the necessary technology for batteries to last the lifetime of the vehicle and reduce the need for replacement.

Life Cycle Assessment: According to Tesla's life cycle assessment, the Model 3 has lower lifetime emissions than a vehicle with an internal combustion engine after driving 5,340 miles. According to a Wall Street Journal study, initial production generates 65% more emissions than the Toyota RAV4 but emissions are roughly the same at 20,600 miles. At 100,000 miles RAV4 has 77% more emissions. By 200,000 miles, the RAV4 has generated 78 tons of carbon emissions while Model 3 has generated 36 tons (1. Gold, 2021). Furthermore, Tesla products have the highest energy efficiency of any electric vehicle on the market and one of the lowest water intensities required for production. The energy efficiency is especially pertinent as electricity mixes of fossil fuels and renewables vary across the United States and the globe. According to company data, despite a Tesla product being charged from natural gas or coal sources, the emissions equivalent is an internal combustion engine that has 59 miles per gallon. The organization is optimistic that as more renewables are introduced to the grid the maximum benefits of their products will be realized.

While Tesla markets itself as a "green" company, several critics have identified inconsistencies.

Greenhouse Gas Reporting: Tesla is among 15 percent of the world's largest companies that do not report the organization's total greenhouse gas emissions or future reduction goals. American automakers Ford and General Motors are quite the opposite, stating their total emissions and related goals. As a

result, this calls into question the transparency of Tesla's data collection, emissions, and environmental claims (2. Bansal, 2021).

Lawsuits for Environmental Violations: The company has been subject to multiple environmental violations and investigations. In 2018, the Bay Area Air Quality Management District fined Tesla \$139,800 for nitrous oxide pollution at the Fremont, California facility. More recently, the company is under investigation by the EPA for failing to disclose proof of compliance with Clean Air Act standards related to the application of surface coating on their vehicles. In Germany, authorities fined Tesla \$14.5 million for failing to uphold battery retrieval and recycling obligations (3. Kolodny, 2021).

Material Sourcing for Batteries: A large pollution and greenhouse gas emissions source associated with electric vehicles is the mining of the battery materials cobalt, lithium, and nickel. The energy intensive and destructive process produces hazardous tailings and slags. If not disposed of properly, the waste can infiltrate groundwater resources and result in toxic levels of heavy metals and chemicals. Currently, a lithium mine is proposed in Nevada, however local Native American tribes are protesting its construction (4. Penn, 2021).

Bitcoin: Elon Musk announced the company would begin accepting bitcoin as a form of payment and has invested \$1.5 billion worth of the currency in January 2021 (5. Bambrough, 2021). With annual energy consumption equal to Sweden's. In China alone, which accounts for 70 percent of bitcoin mining activities, the sector consumes 5.4% of the country's total annual energy production with each unit emitting 10.77 kg of carbon per USD. By 2024, researchers project the industry to emit 130.50 million tons annually (6. Jiang, 2021). Backlash over the associated greenhouse gas emissions resulted in Musk withdrawing the announcement two months later. However, in July of 2021 he alluded to the possibility of accepting the currency once again (Bambrough, 2021).

4.2 Social Impact

Tesla's Annual Impact Report states that the company is committed to building a safe, fair, and inclusive culture for all employees, and they are working to accomplish this through Environmental, Health, and Safety (EHS) practices, comprehensive employee benefits, fostering a culture of diversity and inclusion, supporting workforce development, and engaging in the communities where their factories exist. The company also added a declaration in support of the United Nation's Universal Declaration for Human Rights (UDHR) to the 2020 Impact Report, stating that they expect all suppliers to comply with and promote dignity, respect, and equality for all people.

The company has implemented several programs to ensure a safe and successful work environment for its employees. Take Charge is an EHS program that is designed to engage all employees in building a safety culture by asking them to submit ideas on how to improve operations for environmental, health, safety, or security at Tesla. There were 1.3million reported engagements in 2020 that included safety suggestions, inspections, audits, and safety trainings for employees. In addition to safety practices, Tesla continuously works to provide a competitive wage for its employees, benchmarking against national averages in order to remain competitive. Tesla's average wage for US manufacturing roles in the company was \$21.14/hr (with benefits and equity), while the average wage for manufacturing roles in the US is \$16.73/hr. In an effort to promote from within, Tesla also sends out weekly emails with internal job openings and annual evaluations are conducted to survey for potential promotions.

Employee benefits at Tesla include no-cost contribution medical, dental, and vision insurance, confidential counseling services, company-paid life and disability insurance, financial assistance for temporary housing loss, back up child and elder care, infertility assistance, transgender benefits, and resources for families with children who have cognitive impairments.

Tesla offers several workforce development and education programs in the communities where it operates, including Tesla START, the Manufacturing Development Program, and Tool & Die Apprenticeship. That are necessary for a career at Tesla. The Manufacturing Development Program is a two-year cooperation with community colleges in Nevada, New York, Texas, and California, in which high school graduates are hired by Tesla and complete coursework at those colleges while working for the company. Tesla's Tool and Die Apprenticeship operates in a similar way, offering on-the-job training and classroom learning through partnerships with community colleges in California, Nevada, and Michigan.

The inaugural Diversity, Equity and Inclusion (DEI) report was published in December 2020 to provide transparency through results reporting, add accountability measures for internal teams, and focus on creating new solutions and principles in the DEI space. Per the 2020 report, Tesla considers itself a majority-minority company with 61% of their workforce coming from underrepresented communities. Their leadership teams are primarily White, though Tesla mentions the fact that 0.3% of their workforce is in leadership and 33% of their directors are from minority communities. According to the 2020 Impact Report, many 2021 initiatives are focused on increasing Black and Latinx representation within the company. Women are also underrepresented in the company, which Tesla acknowledges and admits that work needs to be done, noting that increasing representation at all levels is a top priority.

Tesla's DEI Roadmap highlights several initiatives for the future, including a DEI Recruiting Summit for scouting top talent from underrepresented communities, unconscious bias training for recruiters, expansion of an internship program and military fellowship program to attract diverse talent, and quarterly diversity data reviews to ensure that trends are noticed early and remedied across the company.

Notably, Tesla states that employees have "the right to freely discuss their wages, benefits, and terms and conditions of employment and to raise complaints internally or externally." Recent public complaints include several lawsuits alleging that Tesla supports a racist working environment, with one recent settlement requiring Tesla to pay \$137m to a former employee who faced a hostile environment during his tenure at the company.

5. Financial Analysis and Success Indicators

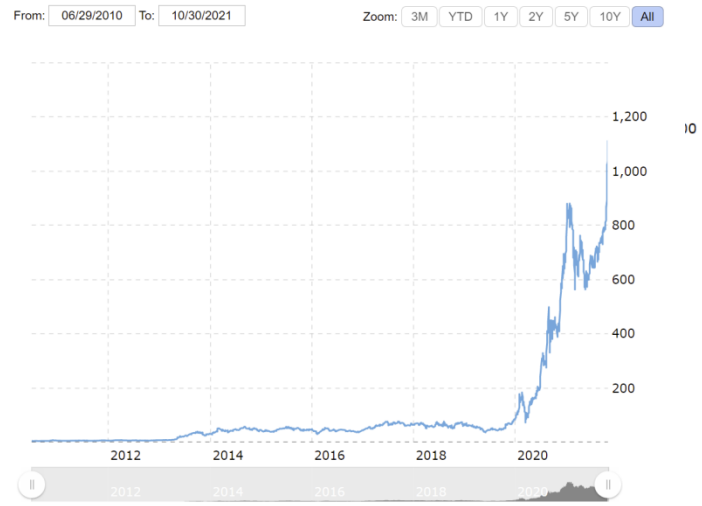
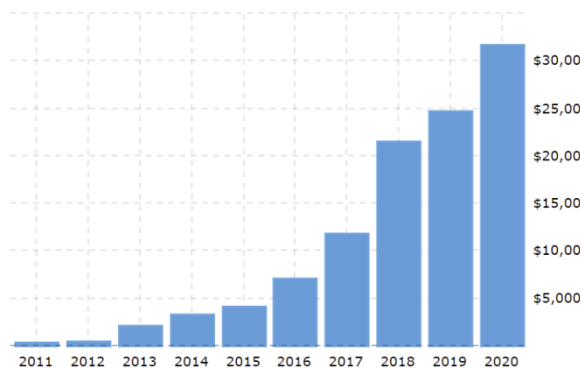
5.1 Financial Analysis

Tesla has had a rocky road to success. The company went from humble revenues of \$14 million in 2008, to being a month away from bankruptcy during the rollout of the Model 3 in 2018 (1. Kolodny, 2020). Tesla has only begun to turn a profit of \$721 million in 2020 as their revenue swelled to \$31 billion with the help of \$1.58 billion of revenue from automotive emission regulatory credit sales (2. Green, 2021).

Tesla Stock price, 2010-2021

TSLA - Revenue

Annual Values (Millions of US \$)

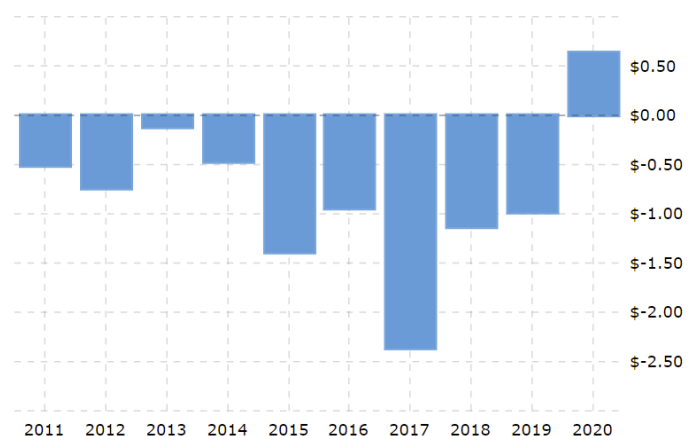


Tesla's stock price follows a similar rocky road (which is hard to see because of the recent price spike) followed by a rise in the stock price starting around January of 2020 that has over about a year and has led Tesla to a market cap of over \$1 trillion. An increase that allowed them to be included into the S&P 500 in December of 2020. It's worth noting, however, that Tesla's stock price and market cap have 'funky' properties, Tesla is worth more than the next seven automakers combined. Its P/E ratio is currently over 450: when compared to the average of 30 for the S&P 500 (3. Multpl, 2021) it is clear this company is valued for more than just its performance, which as we saw above is impressive, but has only recently become profitable.

Looking over what most analysts and those following Tesla have to say, the general view is simply this: If you look at Tesla as an auto-manufacturer, their price-to-earnings and market cap do not make sense. If you look at Tesla as a software and tech company (owned by Elon Musk) that is creating the industries shaping the 21st century, the numbers are not as shocking.

TSLA - EPS - Earnings per Share

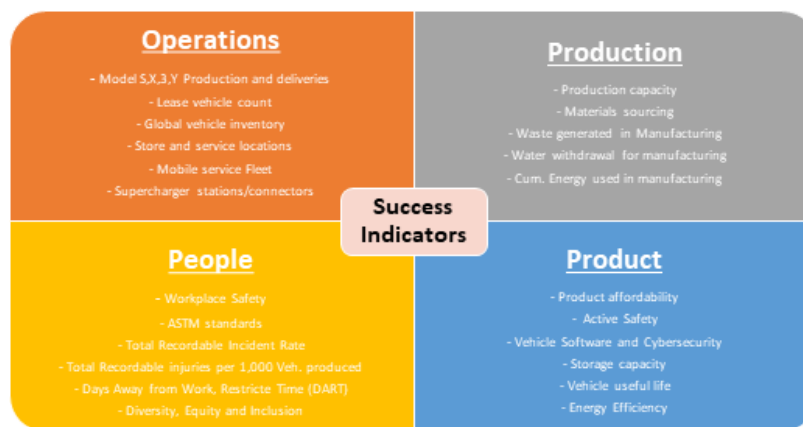
Annual Values (US \$)



Tesla	2020	2015	2009
Current Ratio	0.86	0.99	1.75
Debt/Equity	0.509	2.488	-0.0043
Inventory Turnover	6.07	2.444	4.41
ROA	1.653	-11.015	-42.737
ROE	3.736	-82.002	21.986
Net Profit Margin	2.188	-21.964	-49.793

Rank	Name	Market Cap	Price Today	Price (30 days)	Country
1	 Tesla TSLA	\$1.197 T	\$1,192 7.05%		us USA
2	 Toyota TM	\$251.28 B	\$179.75 1.80%		JP Japan
3	 Volkswagen VOW3.DE	\$142.63 B	\$225.07 0.43%		DE Germany
4	 BYD 002594.SZ	\$134.95 B	\$47.17 -2.99%		CN China
5	 Daimler MBX1.DE	\$107.58 B	\$100.57 1.45%		DE Germany
6	 General Motors GM	\$80.31 B	\$55.32 1.64%		us USA
7	 Great Wall Motors 807603.SS	\$76.37 B	\$10.25 -3.53%		CN China
8	 Ford F	\$71.50 B	\$17.89 4.76%		us USA

5.2 Success Indicators



(Tesla, 2020)

Operations: High efficiency in operations and extended distribution networks have allowed Tesla to become the leader in the EV market. Building capacity to supply and meet demand is crucial for sustaining growth in the long term. Some key operating indicators include production and deliveries, global vehicle inventory, store and service locations, mobile service fleet, and supercharger stations and connectors distribution.

Production: Currently, Tesla is trying to stretch their EV production by increasing capacity in their California and Texas factories, and opening a new facility in Berlin, Germany. The company has set a diverse set of metrics to maximize production and streamline its processes throughout its entire supply chain. From materials sourcing to product deliveries, Tesla oversees each production stage to ensure low environmental impact.

People: Creating a working environment that increases productivity and employee engagement has been a priority for Tesla. Safety measures and compliance with ASTM standards have significantly reduced incident rates and injuries. Employee attraction, retention, and development through diversity and engagement programs reinforce Tesla's commitment to Environmental Social Governance (ESG).

Product: Most of Tesla's efforts are focused on increasing the efficiency and safety of their vehicles. However, user experience and affordability are becoming increasingly valuable aspects for customers. Full Self Driving (FSD) and infotainment features set apart Tesla's models from their competitors,

providing a uniquely immersive experience while also making their vehicles accessible to more customers. (Tesla, 2020)

6. Analysis, Recommendations, and Conclusion

Overall, Tesla has propelled itself and the electric vehicle industry forward through innovation, risk-taking, and the cult of personality centered around Elon Musk. With the purpose of accelerating the world's transition to sustainable energy without compromising quality and convenience, the organization disrupted the stigmas surrounding the industry and created a "blue ocean" focused on electrifying the traditional transportation sector. Sharing their breakthroughs in battery technology and product design through open sourcing patents, Tesla has put actions behind its words.

Despite their well-intentions, the organization has faltered in their social and environmental impact disclosure. Specifically, we recommend that Tesla increase the transparency in reporting their total greenhouse gas emissions, and to prioritize the implementation of DEI initiatives.

To drive systems change, we recommend that Tesla increase its corporate political activism, specifically targeting a carbon tax to reflect the true price of conventional fuels, greater tax incentives for EV purchases, and infrastructure projects centered around EV charging capacity. We also recommend that Tesla divest from Bitcoin as it is greenhouse gas intensive and does not coincide with the organization's purpose of accelerating the world's transition to sustainable energy.

With their past performance, innovation and future plans, even if they do not implement our incredibly well researched recommendations, we expect Tesla to continue to be successful in the future.

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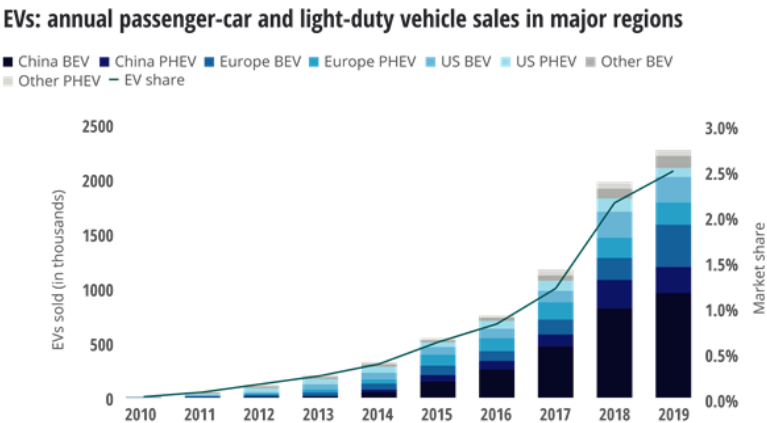
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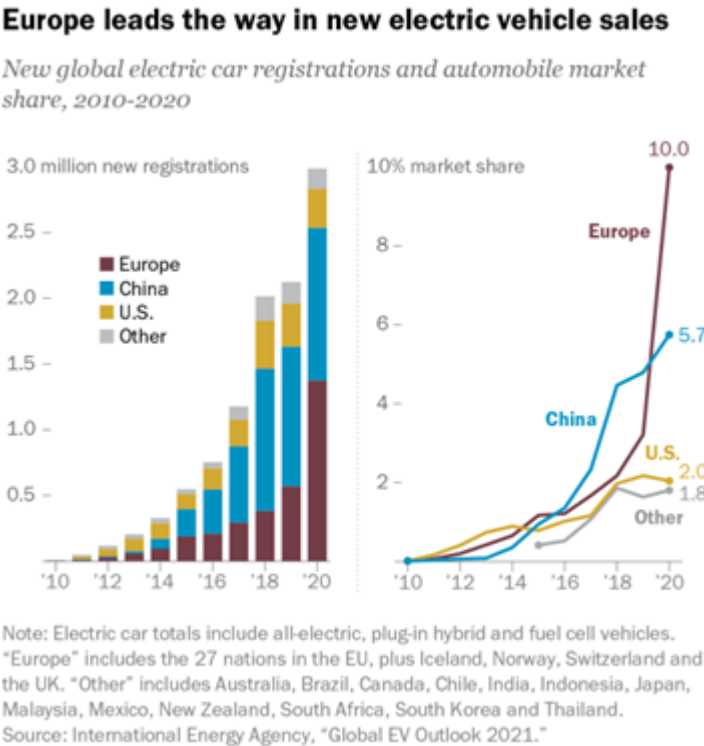
8. Appendix



Source: Deloitte analysis, IHS Markit, EV-volumes.com²

Figure 1: Deloitte Insights | deloitte.com/insights

(1. Deloitte, 2021)



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Figure 2:

(2. DeSilver, 2021)

Electric vehicles make up large share of new car sales in Northern Europe

New electric cars as share of automobile sales, globally and in selected countries, 2020



Source: International Energy Agency, "Global EV Outlook 2021."

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Figure 3:

(DeSilver, 2021)

Consumer priorities for EV adoption, 2018 and 2020

Greater concerns are shown in orange.

2020 Global Auto Consumer Study												
	FRANCE		GERMANY		ITALY		UK		CHINA		US	
In your opinion, what is the greatest concern regarding all battery-powered electric vehicles?	2018	2020	2018	2020	2018	2020	2018	2020	2018	2020	2018	2020
Driving range	31%	28%	35%	33%	4%	27%	26%	22%	25%	22%	24%	25%
Cost/price premium	32%	22%	22%	15%	19%	13%	24%	16%	9%	12%	26%	18%
Time required to charge	11%	15%	11%	14%	18%	16%	13%	16%	12%	15%	10%	14%
Lack of electric vehicle charging infrastructure	16%	22%	20%	25%	44%	32%	22%	33%	18%	20%	22%	29%
Safety concerns with battery technology	4%	11%	5%	10%	7%	10%	6%	12%	22%	31%	8%	13%
Others	6%	2%	7%	3%	8%	2%	9%	1%	14%	0%	10%	1%
Total	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%
Sample size	1,083	1,266	1,287	3,002	1,048	1,274	965	1,264	1,606	3,019	1,513	3,006

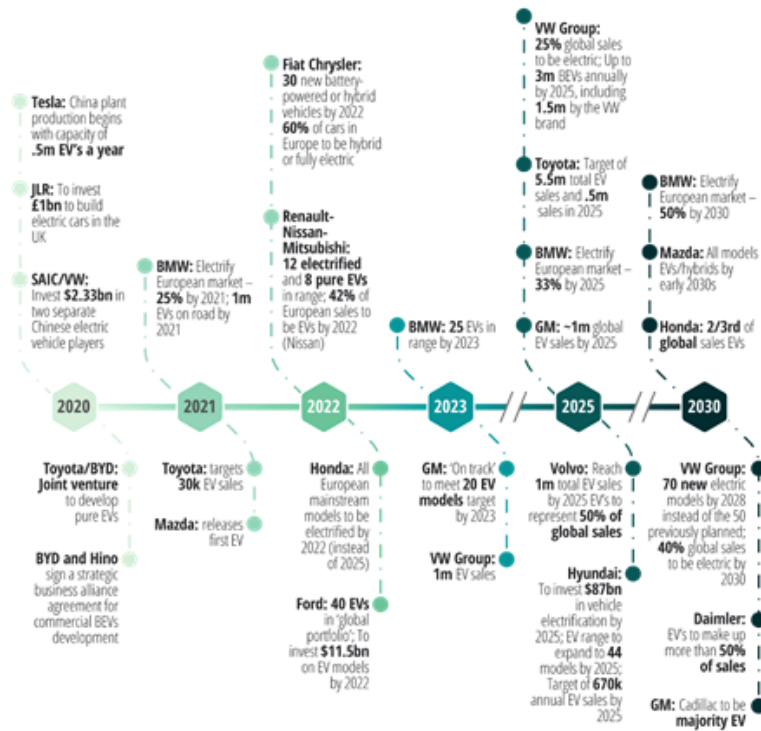
Source: Deloitte Global Auto Consumer Study[®]

Deloitte Insights | deloitte.com/insights

Figure 4:

(3. McKinsey, 2021)

Timeline of strategic OEM targets for EVs



Source: Deloitte analysis²⁸

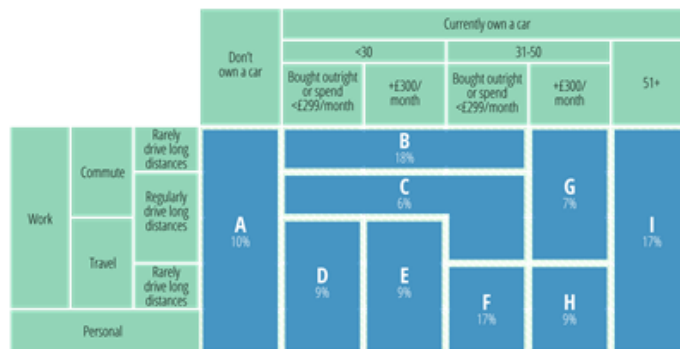
Figure 5:

Deloitte Insights | deloitte.com/insights

(Deloitte, 2021)

The nine consumer segments of the United Kingdom automotive market

% reflect overall size of the segment



* Using the annual average GBP/USD exchange rate for 2019, <£299/month = <\$382/month and +£300/month = +\$383/month

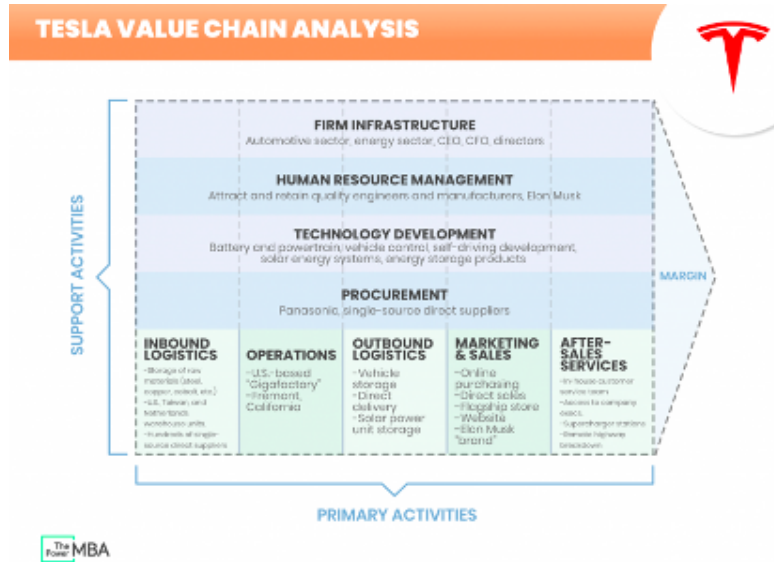
Source: Deloitte analysis²⁹

Figure 6:

Deloitte Insights | deloitte.com/insights

(Deloitte, 2021)

Figure 7:



(4. The Power MBA, n.d.)

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